B. Big Vova

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1300 | Tags: brute force, greedy, math, number theory | https://codeforces.com/problemset/problem/1407/B

Alexander is a well-known programmer. Today he decided to finally go out and play football, but with the first hit he left a dent on the new Rolls-Royce of the wealthy businessman Big Vova. Vladimir has recently opened a store on the popular online marketplace "Zmey-Gorynych", and offers Alex a job: if he shows his programming skills by solving a task, he'll work as a cybersecurity specialist. Otherwise, he'll be delivering some doubtful products for the next two years.

You're given n positive integers $a_1, a_2, \ldots, a_n$. Using each of them **exactly at once**, you're to make such sequence $b_1, b_2, \ldots, b_n$ that sequence $c_1, c_2, \ldots, c_n$ is *lexicographically maximal*, where $c_i=GCD(b_1,\ldots,b_i)$ - the greatest common divisor of the first i elements of b.

Alexander is really afraid of the conditions of this simple task, so he asks you to solve it.

A sequence a is lexicographically smaller than a sequence b if and only if one of the following holds:

- a is a prefix of b, but $a \neq b$;
- in the first position where a and b differ, the sequence a has a smaller element than the corresponding element in b.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^3$). Description of the test cases follows.

The first line of each test case contains a single integer n ($1 \le n \le 10^3$) — the length of the sequence a.

The second line of each test case contains n integers $a_1,\ldots,a_n$ ($1 \le a_i \le 10^3$) — the sequence a.

It is guaranteed that the sum of n over all test cases does not exceed 10^3.

Output

For each test case output the answer in a single line — the desired sequence b. If there are multiple answers, print any.

Example

Input

```
7
2
2 5
4
1 8 2 3
3
3 8 9
5
64 25 75 100 50
1
42
6
96 128 88 80 52 7
5
2 4 8 16 17
```

Output

```
5 2
8 2 1 3
9 3 8
100 50 25 75 64
42
128 96 80 88 52 7
17 2 4 8 16
```

Note

In the first test case of the example, there are only two possible permutations b — [2, 5] and [5, 2]: for the first one c=[2, 1], for the second one c=[5, 1].

In the third test case of the example, number 9 should be the first in b, and $GCD(9, 3)=3$, $GCD(9, 8)=1$, so the second number of b should be 3.

In the seventh test case of the example, first four numbers pairwise have a common divisor (a power of two), but none of them can be the first in the optimal permutation b.